Network Swiss Army Knife (NSAK)

Bachelor Thesis – Integrating Agentic AI for Network Reconnaissance

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Introduction

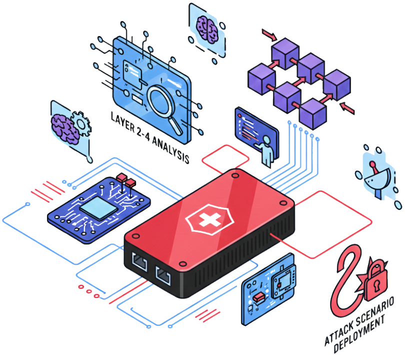
Starting Point: The NSAK Framework

Established in the predecessor project (Project 2)

A modular, open-source network security framework

Built around four reusable resource types:

- **Device** *execution host at a network position*
- **Environment** *target network topology*
- **Scenario** *a red/blue team use case*
- **Drill** *reusable building block*



Hypothesis and Research Questions

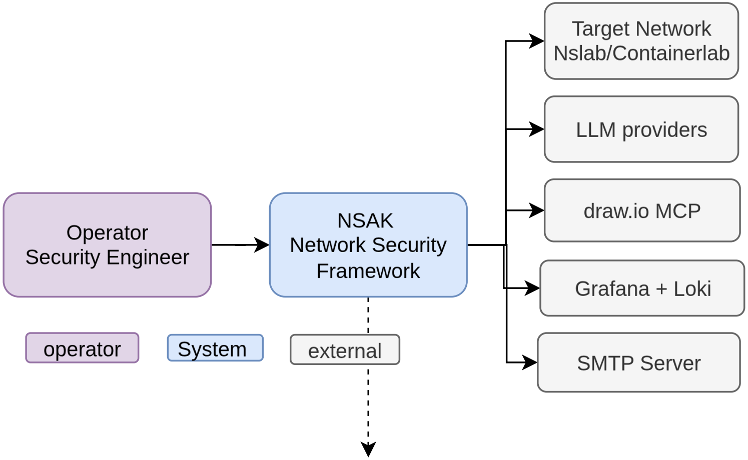
From static scenarios to agentic AI

Hypothesis

Integrating **agentic AI** into NSAK improves **adaptability, flexibility** and **automation** over static scenarios, while **lowering the operator's required expertise** to interpret results and plan follow-up steps.

Layered Architecture

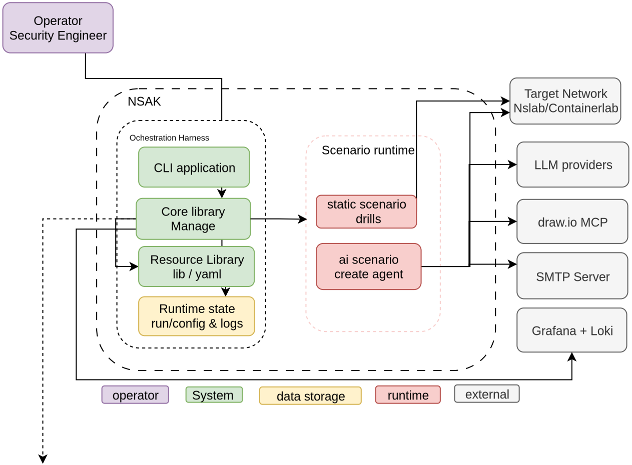
system-diagram



Background and Concepts

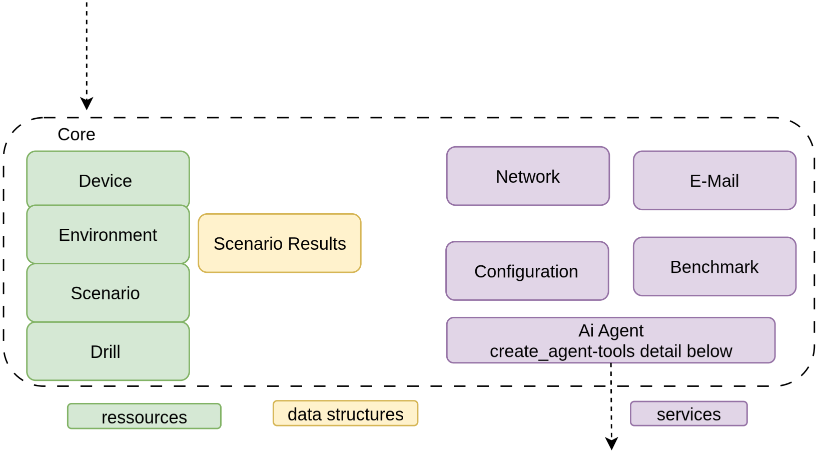
Layered Architecture

Lean Component Diagram



Layered Architecture

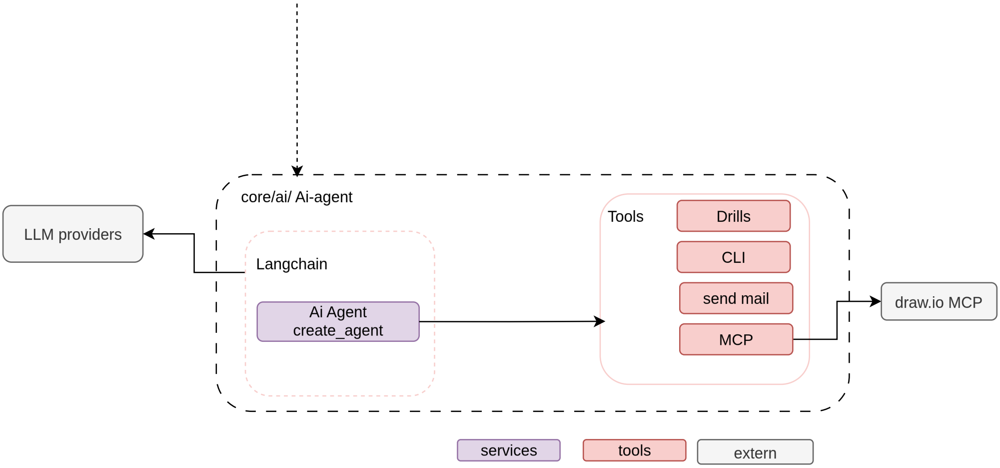
Core Diagram and Agent



Implementation

Layered Architecture

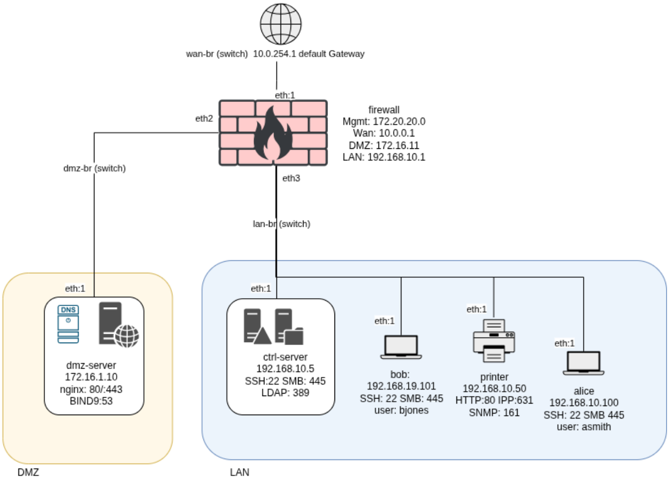
Core Diagram and Agent



Evaluation Setup

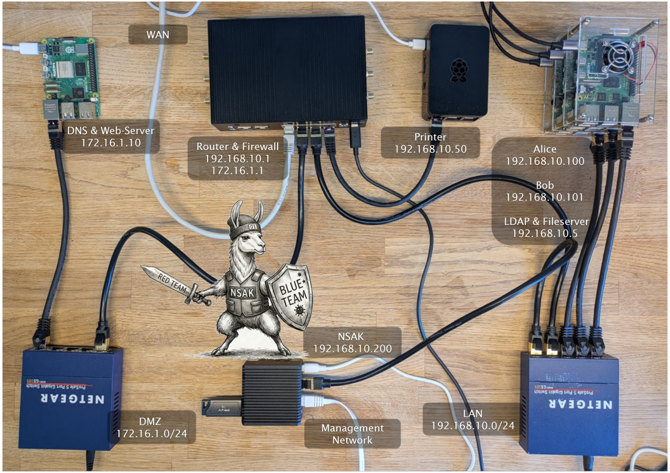
Reproducible Test Environment

A virtual enterprise network built with Containerlab



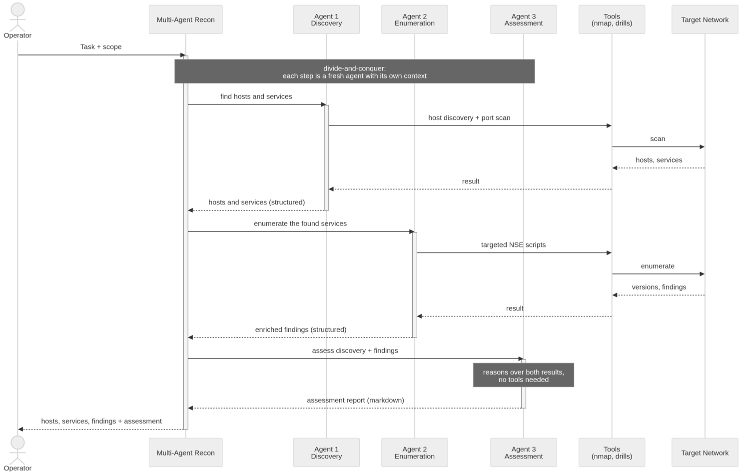
Physical Lab

A virtual enterprise network built with Containerlab



Multi-Agent Reconnaissance Pipeline

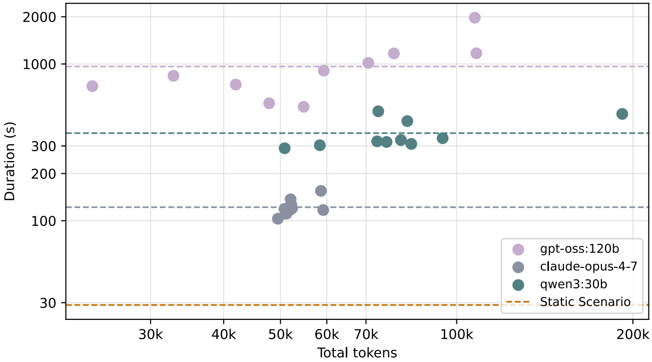
Discovery → enumeration → assessment, each a fresh agent



Results

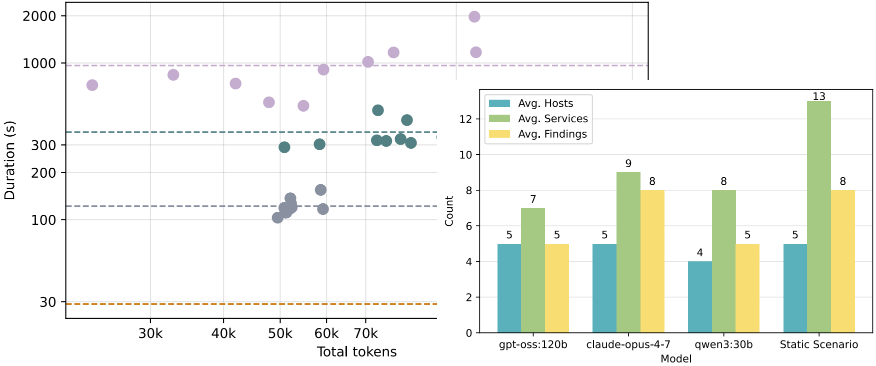
Containerlab – Multi-Agent Benchmark

Containerlab Findings



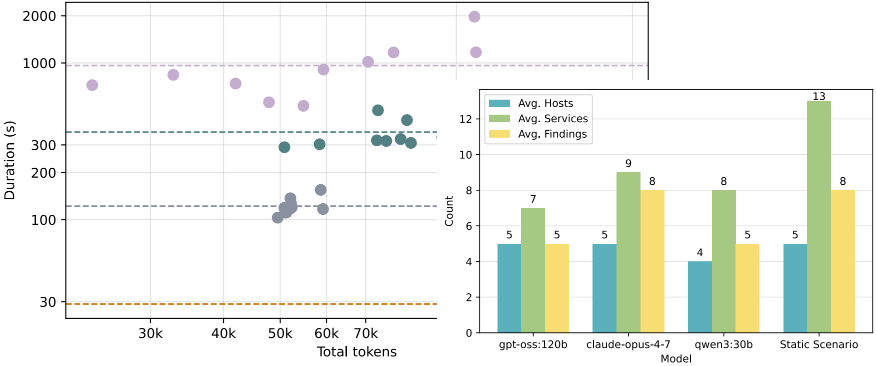
Containerlab – Multi-Agent Benchmark

Containerlab Findings



Containerlab – Multi-Agent Benchmark

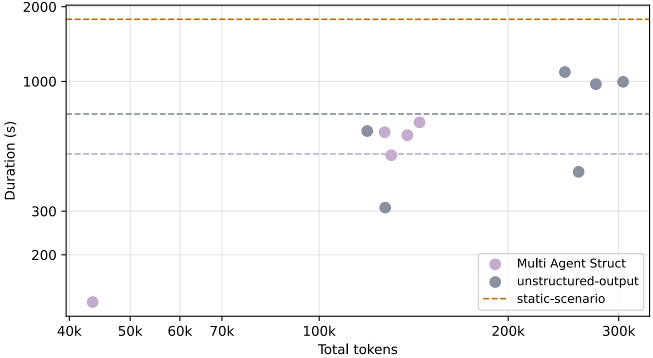
Containerlab Findings



- Decomposition: `qwen3:30b` drops tokens –68% and duration –30%.
- `claude-opus-4-7` keeps identical discovery quality (slight +13% token overhead).

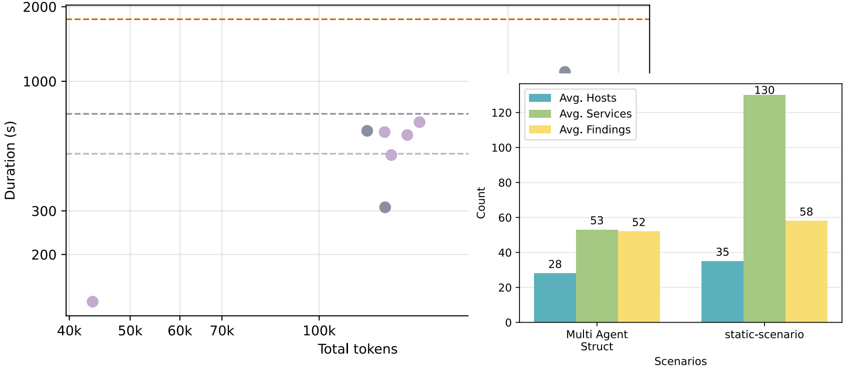
NS Lab – Multi-Agent Benchmark

Containerlab Findings



NS Lab – Multi-Agent Benchmark

Containerlab Findings



Discussion and Conclusion

Key Findings

Takeaways across both environments

What works

- Even smaller models can autonomously run **standard red-team reconnaissance**
- **Multi-agent decomposition** helps smaller models (–68% tokens for `qwen3:30b`)
- AI delivers **depth**: assessment, prioritized findings, remediation

Key Findings

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What limits it

- **High failure rate** – best-case subset
- **Hallucinations**: fabricated CVEs, honeypots
- **Model tier matters**: `qwen3:30b` unusable
- Manual qualitative scoring

Recommendations and Future Work

Recommendations

- **Match model tier to task** – frontier model for autonomous recon
- **Multi-agent** for smaller / self-hosted models
- **Hybrid pipeline:** static breadth + AI depth (+ RAG)
- **Human-in-the-loop** + retry on incomplete runs

Future work

- **Dynamic tool calling** – expose only relevant tools
- **Guardrails & sandboxing** – scope restriction for production
- Robust **HITL middleware**
- **Broader attack-chain** coverage beyond reconnaissance

Demo: Agentic Reconnaissance in Action

Questions?